NIHR | National Institute for Health and Care Research

Reference number	NIHR207992
Programme	Research for Patient Benefit
Call	RfPB Competition 53 Stage 1
Lead Applicant	Dr Richard Ma
Host organisation	NHS BRISTOL, NORTH SOMERSET AND SOUTH GLOUCESTERSHIRE INTEGRATED CARE BOARD
Research Title	Optimising primary health care for people living with HIV– using a Person Based Approach to co-design a model of care for UK general practice
Start Date	03/03/2025
Grant Duration (Months)	18
End Date	02/09/2026
Estimated Research Cost	£180,401.32
Estimated NHS Support & Treatment costs	£.00
Research Type	Primary Research

Lead Applicant Details

Lead Applicant

Full Name	Dr Richard Ma	**Position**	Clinical Lecturer
ORCID iD	0000-0001-6877-2785	**Institution**	University of Bristol

Specify your (lead applicant) role in the research
As lead applicant, my role includes: • Conception and design of research, methodological input and analysis • Identifying and co-ordination of collaborators • Communication, planning and engaging stakeholders with protocol and grant application • Data curation including applying for access to database of electronic health records, "cleaning" of data and software coding • Project management and administration • Co-supervision of research assistant • Drafting, critically reviewing and approving final version of outputs for conference presentations and publication • Planning and co-ordinating dissemination activities including social media output and engagement events Towards the end of the project, I will be applying for a further grant for the next steps

% FTE commitment	10

Joint Lead Applicant

Full Name	Dr Jeremy Horwood	**Institution**	University of Bristol
ORCID iD	0000-0001-7092-4960	**Position**	Professor in Social Science and Health

Justification for Joint Lead Applicant and role in research
RM is at the immediately postdoctoral stage of his career so needs support to prepare for the next stage of his academic journey. JH is RM's mentor and has experience and expertise to support RM with this application. JH co-leads the Behavioural and Qualitative Science Team for NIHR ARC West and is the co-director of the Sexual Health Improvement Programme (SHIP) Health Integration Team. His research interests include users' experiences of health and health services, primary care and sexual health. His research experience and interest would be helpful to support RM's role in this application and a springboard to future grant applications and transition into an independent researcher. JH's role includes • conceptualisation of research, goals and aims • methodological support • support with stakeholder engagement • co-supervision of research assistant • preparation, critical review and approval of outputs • support with dissemination activities JH will be involved with further grant applications resulting from this project.

% FTE	2

Research Team Details

Co-Applicant	**Organisation**	**Specify Co-Applicant role in research**	**% FTE**
Professor Jaime Vera Rojas	Brighton and Sussex Medical School	JVR brings clinical expertise as well as a track record in NIHR-funded research in cognitive and physical impairment, and management of comorbidities in older PLWHIV.	2.5

Co-Applicant	Organisation	Specify Co-Applicant role in research	% FTE
Dr Maria Theresa Redaniel	University of Bristol	MTR has expertise in using electronic health records in mixed methods evaluations, health services research and epidemiology, and will help support the senior research associate.	2.5
Ms Angelina Namiba	4M Mentor Mothers Network CIC	AN has lived experience of HIV and represents an HIV charity. She has track record of involvement in NIHR-funded research and will co-lead on PPI.	1.0
Ms Helen Bolton	University of Bristol	Ms Bolton will provide advice and hands-on support for communicating about research progress and dissemination such as press releases, news stories and social media content.	2.5
Dr Werner Leber	Queen Mary University of London	WL is an academic GP with expertise in cluster randomised controlled trials and MRC complex implementation studies of HIV testing in primary care.	1.5
Mr Tom Hayes-Isaacs	SAVING LIVES	THI has lived experience of HIV and has been involved in HIV advocacy, research, policy and clinical standards. He will co-lead PPI work with AN.	1.0

COI declaration
Please declare any conflicts or potential conflicts of interest that you or your co-applicants may have in undertaking this research, including any relevant personal, non-personal and commercial interest that could be perceived as a conflict of interest.

Dr Richard Ma is a trustee of Sussex Beacon which is an HIV charity based in Brighton and has received funds from Gilead Sciences to attend a conference and ViiV healthcare for a round table discussion on HIV testing in general practice.

Dr Werner Leber has received funds from Gilead Sciences.

Dr Hilary Curtis is funded by Gilead Sciences for consultancy work.

Dr Jane Nicholls has received grants from and done consultancy work for Gilead Sciences.

Agreement to terms and conditions.	☒Confirmed

Plain English Summary of Research

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Aim(s) of the Research

We want to develop a way in which people living with HIV (PLWHIV) can see their general practitioners (GPs) regularly for health checks and help them manage their HIV and other conditions to keep them healthy as they get older.

Background to the Research

PLWHIV now have a near-normal lifespan due to effective treatment for HIV. There are now double the number of PLWHIV over the age of 50 than there were 10 years ago. They are more likely to develop mental health issues, heart problems, diabetes and kidney problems compared with people without HIV. This is why they need extra health checks and care from their GPs.

There isn't much guidance on what GPs should do to support PLWHIV, nor what works best. Not everyone tells their GP about their HIV diagnosis nor sees them due to stigma and past poor experiences with healthcare.

Design and Methods Used

The research uses the Person-Based Approach, where the input of PLWHIV is central to the planning and carrying out the research. We aim to develop better ways of caring for PLWHIV that reflects what is wanted and needed by patients and care providers.

There are three parts to our research:

1. We will review recent studies on different models of care (ways of providing healthcare) and get feedback from PLWHIV.
2. We will use a database of anonymised GP records to examine if PLWHIV are getting health checks from their GPs.
3. Using the results of the review of studies, a steering group (made up of PLWHIV, GPs, HIV specialists, and researchers) will design an initial model of care for PLWHIV. We will hold focus group with other PLWHIV and health care professionals to get their feedback on the model we have developed.

The steering group will make changes to the model from feedback. A revised model will be tested in "the real world" in a future study. The findings will provide evidence for clinical guidance to improve the health care of PLWHIV so that they can live longer and healthier lives.

Patient and Public Involvement (PPI)

Two people with lived experience of HIV have helped to write the research proposal and will be part of the study team throughout the project. They will help design study materials, co-lead the steering group, review results, write and help share research findings.

Dissemination

When we have the results of the study, we will share them with the public (including PLWHIV) and health workers by publishing in research journals and talking at conferences. We will work with our PPI and communications team to write news stories for our website to share via social media and write press releases e.g. for newspapers and TV.

Research Plan

Research Plan

What is the problem being addressed?

Antiretroviral therapy (ART) introduced in the 1990s has transformed human immunodeficiency virus (HIV) disease from a life-threatening condition leading to AIDS (Acquired Immuno-Deficiency Syndrome) and premature death, to a manageable long-term condition. Life expectancy for people living with HIV (PLWHIV) is approaching that of the general population.(1)

PLWHIV are experiencing ageing and its effects.(2) Compared to people without HIV, PLWHIV are more likely to have worse physical and mental health, increased risks of cardiovascular disease, kidney disease, diabetes and frailty.(3-5) Prevalence of comorbidities (i.e. more than one disease coexisting in one person) among PLWHIV in the UK such as cardiovascular disease and mental health conditions are also increasing, in addition to increases in deprivation.(6, 7) Comorbidities and frailty have serious consequences for PLWHIV and the UK National Health Service (NHS) in terms of reduced quality of life, unemployment, depression, increased use of medications and healthcare services.(8-12)

In the UK NHS, general practice provides most of the first-contact primary health care and preventative care (e.g. health promotion, disease screening and vaccination). At least 96% of the UK population are registered with a general practitioner (GP), and on average, consult their GP five times a year.(13, 14) Multimorbidity (more than two diseases in one person) is common in the general population; one study reported 58% of patients attending general practices had multimorbidity, but they accounted for 78% of all consultations.(15)

HIV clinic staff might not have the expertise to deliver care to an ageing cohort of PLWHIV with comorbidities, mental health needs and frailty compared with GP teams. This project will explore the role of general practice in the long-term care for PLWHIV to help them receive the right care, in the right place, that is right for them, and lead longer and healthier lives.

Why is this research important?

New HIV diagnoses in England rose by 22% from 3,118 in 2021 to 3,805 in 2022. There were 94,397 PLWHIV accessing care in England in 2022, a rise of 3% from 91,368 in 2021. Due to successful management of HIV, the proportion of PLWHIV aged over 50 years have almost doubled in 10 years from 27% in 2012 to 50% in 2022. (16)

Over a quarter of adults in England are estimated to have multimorbidity, and is expected to double between 2015 and 2035.(17) People with long-term conditions use more health services, accounting for 70% of the total health and social care spend in England alone.(18) Despite this, levels of unmet need are high, particularly for support from community services to manage multiple conditions.(19)

Investment in general practice to provide capacity and designing integrated healthcare around patient needs have been the driving principles behind *The NHS Five Year Forward View* and *The NHS Long Term Plan*.(20, 21) This research proposal aligns with these priorities by generating evidence on how general practice can contribute effectively to the long-term care of PLWHIV, so that they benefit from longer and healthier lives, and health service resources are used effectively .

This study will contribute to evidence-based clinical guidance on the care of PLWHIV in general practice to be implemented within five years of this project to benefit the health and care of an ageing cohort of PLWHIV.

Review of existing evidence

Despite the potential contribution of general practice to deliver long-term and preventative care for PLWHIV, there is little clinical guidance on what this might look like. The British HIV Association (BHIVA) published a scoping review in 2017 on different care models (ways of delivering care) to PLWHIV.(22) It identified three models used to manage long-term conditions that might be relevant to HIV: current standard specialist model, shared-care and collaborative-care models.

In the current standard model, the HIV clinic manages all issues related to HIV, and any non-HIV conditions are referred back to GPs. A shared-care model has agreed care protocols between partners to support the co-ordination of care.(23) There were few

examples where GPs managed most aspects of HIV care including drug monitoring, but the quality of evidence was poor; there were no consistent association with improved clinical outcomes, cost effectiveness or acceptability.(24) Collaborative-care models originated from mental health services and use case-based management and structured care plans shared with everyone involved in the care of the individual. While they are more fluid and responsive, they require resources to support communication to all health and social care providers. For a very long-term condition such as HIV, it needs to change focus for different support services at different stages of an individual's HIV journey.(22)

For PLWHIV, there are barriers to receiving care from GPs. Poor experiences, whether real or perceived, might make it less likely for PLWHIV to share their diagnosis and attend check-ups. A systematic review reported common factors that are key to improving PLWHIV's experiences of GP care include: perceived competence, communication and continuity.(25)

Before we explore what a good care model might look like, we need to conduct an up-to-date scoping review to map out current practices as some research are now dated. This review needs to expand on the previous BHIVA report to include PLWHIV's experiences of general practice to examine barriers and facilitators to access, what care is already being provided by GPs to PLWHIV, and what the gaps might be. An NIHR-funded feasibility study of a geriatric assessment for PLWHIV in hospital settings is underway.(26)

Research question / aims and objectives

Aims and objectives

To develop a model of care to optimise primary health care for PLWHIV in UK general practice

Proposed research questions:

1. How is primary health care delivered to PLWHIV?
2. Are PLWHIV receiving primary health care from their GPs?
3. What does a "good" model of primary health care for PLWHIV look like?

Objectives

1. To conduct a scoping review to map out primary health care delivered to PLWHIV.
2. Using a database of electronic health records in general practice, examine the proportion of PLWHIV who received primary health care and if this varied among different regions, demographic and socioeconomic groups
3. To work with key partners including PLWHIV, HIV specialists and GPs, to co-design an intervention to optimise primary health care provision for PLWHIV.

Research Team

Our research team reflects a mix of clinical and research expertise; experience in advocacy, commissioning; and track record of public engagement including with PLWHIV.

- Joint lead applicants: Richard Ma, Jeremy Horwood
- Co-applicants: Jaime Vera-Rojas, Maria-Theresa Redaniel, Werner Leber, Angelina Namiba, Tom Hayes-Isaacs, Helen Bolton
- Collaborators: Sarah Denford, Catherine Dodds, Jo Copping, Jane Nichols, Philip Bright, Hilary Curtis
- PPI Advisor: Alisha Newman
- Senior research assistant (SRA - to be recruited)

GP academics RM and WL will provide insight into primary care delivery and access to networks of primary care clinicians for study recruitment. JVR, JN and PB provide the same for HIV specialists. RM analysed electronic health records and focus groups for his PhD research funded by NIHR fellowship.

JH, WL and JVR have a track record of publications and successful NIHR grant applications on care of PLWHIV and HIV testing in general practice.

MTR has expertise in the use of electronic health records in mixed methods evaluations, health services research and epidemiology, will give methodological and statistical advice, and help supervise the senior research associate.

SD has published work using PBA and scoping reviews. She also runs workshops on PBA. She will give expert advice on these methods.

JC is a director of public health and co-leads the local Sexual Health Improvement Programme with JH. CD is a policy researcher and has co-produced a collaborative training resource to reduce HIV stigma in healthcare.

ANa and THI have experience in community advocacy and PPI on NIHR-funded programmes; they will lead the PPI / stakeholder engagement, with advice and training given by ANe.

HB is the communications officer and is responsible for leading on disseminating activities for this project. JN, HC, THI and RM are members of BHIVA external relations committee and will be involved in in disseminating research findings through their networks including conferences.

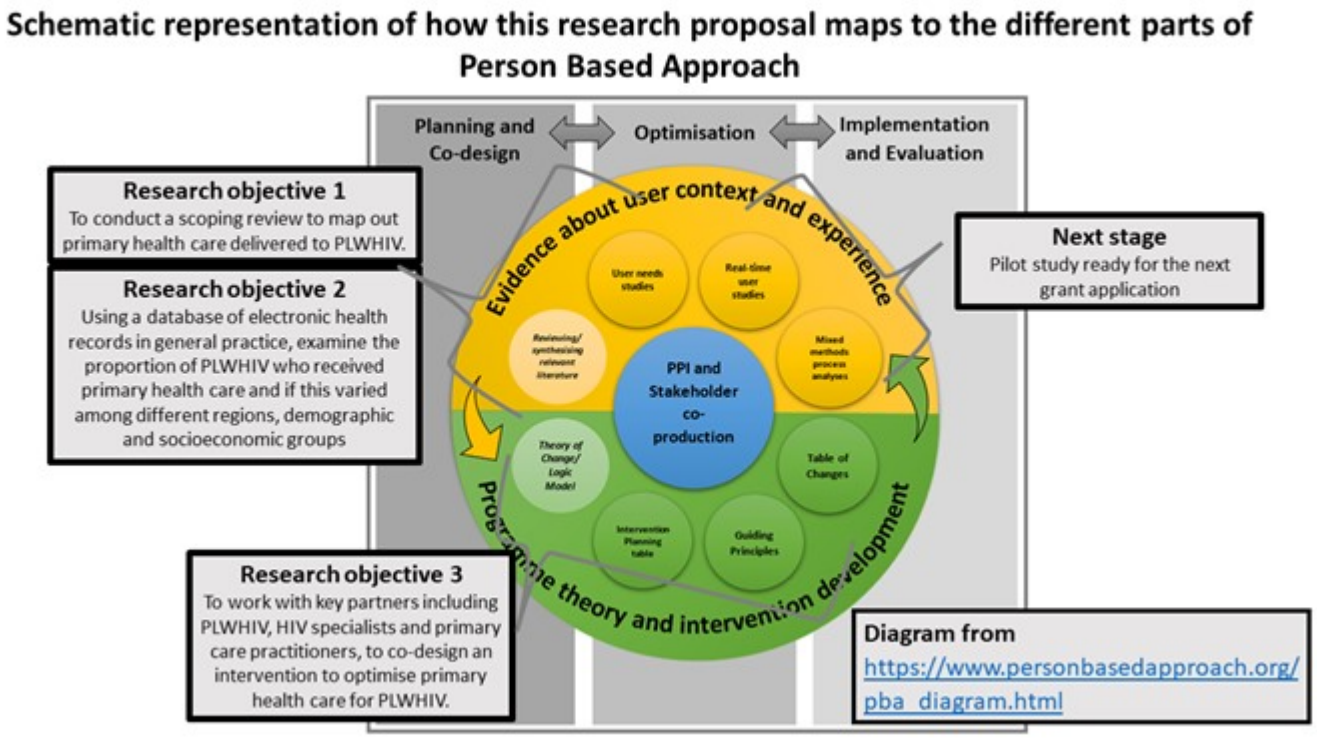
RM, WL, JVR, ANa, THI, JN, PB, JC and HC will also form part of the stakeholder group in addition to recruited members.

Project Plan (including Patient and Public Involvement)

Stakeholder involvement using Person-Based Approach (led by ANa/THI/ANe/SD)

We will use a Person-Based Approach (PBA) throughout all three workstreams to engage with PLWHIV to develop a care model that benefits them (see diagram).(27) Two co-applicants (ANa,THI) are PLWHIV and work in the HIV charity sector; their involvement will ensure PLWHIV influence decision-making at all levels throughout the study.

The PBA offers practical and systematic methods of carrying out many of the key intervention development activities recommended by the Medical Research Council (MRC) funded INDEX guidance for identifying and assessing approaches to developing complex interventions.(28) It has been used to design interventions that demonstrated effectiveness in randomised trials for reducing antibiotic prescribing in primary care. (29, 30) PBA supports meaningful patient and public involvement (PPI) in research. The public and stakeholders are involved in the co-production of an intervention which improves its ownership, applicability and acceptability in practice. The PBA process is transparent as the public and other stakeholders' views are systematically recorded, acknowledged and acted upon.



Stakeholder group

We will convene a stakeholder group to meet every four months online to oversee the three workstreams and lead on the co-design of the intervention (care model) in workstream 3. This group will be co-led by the public co-applicants (ANa, THI) and RM, with support from the PPI Advisor at Bristol University (ANe), with additional members from other PLWHIV (up to four recruited from clinics and peer networks), general practice (GPs and practice nurses) and HIV services (doctors and nurses). Training on PBA will be provided. The activities of this group, including compensation for time and expenses are costed in the budget. A separate Study Steering Group (comprised of the main and co-applicants) will have responsibility for the day-to-day management and delivery of the project.

Equality, diversity and inclusion

PLWHIV are not a homogenous group. Men who have sex with men and Black African people are disproportionately affected; there were 212 trans and gender diverse people seen for HIV care in England in 2022. (16) The number of PLWHIV is higher in metropolitan areas but those outside might have different needs. We will ensure PLWHIV in our stakeholder group not only reflects the demographics most affected by HIV, but also those who might be underrepresented.

Workstream 1 –Scoping Review (months 2 to 9, led by RM/JH/JV/WL/SD/SRA)

Research question 1 - How is primary health care are delivered to PLWHIV?

We will conduct a scoping review to identify how primary health care, such as cardiovascular risk screening, vaccinations and cervical cytology, are delivered to PLWHIV.

This scoping method is preferable to a systematic review because of the range of literature that needs to be explored (quantitative/qualitative research and audits) and broad scope of enquiry. Scoping reviews still require rigorous and transparent methods to ensure the results are trustworthy.(31) We will use a six-stage methodological framework developed by Arksey and O'Malley to: identify the research question; search for relevant studies; select studies; chart the data; collate, summarise, and report the results; and consult with stakeholders to inform or validate study finding.(32)

We will conduct a search of published and grey literature using MEDLINE, EMBASE, NICE Evidence, the Cochrane Collaboration, Google and specialist websites (including, Aidsmap, and World Health Organization), and trial registries such as the International Standard Randomised Controlled Trial Number (ISRCTN). The review will be summarised as: evaluations of care models delivered in primary care for PLWHIV, ethnographic summaries of experiences of primary care and scope of primary health care delivered in specialist settings. The findings will inform the co-design of an intervention for Workstream 3.

To save research costs and improve efficiency, there is capacity to make a start on this workstream at an early stage.

Workstream 2: analysis of a database of electronic health records from general practice (Months 1 to 12, led by RM/MTR/JV/WL/SRA

Research question 2- Are PLWHIV receiving primary health care from their GPs?

Database

We will use the Clinical Practice Research Datalink (CPRD) of anonymised electronic health records. The CPRD Aurum version provides records of 19 million patients from 738 GP practices in the UK.(33) Analysis using CPRD offers the best estimates of current levels of primary health care for PLWHIV and the findings are likely to be generalisable to the UK population. Data will be anonymised, stored in a secure “SafeHaven” folder and access will be limited to the Workstream 2 research team (MTR and RM).

Sample size

It is difficult to prescribe a minimum sample size to give robust estimates for the study. HIV is relatively uncommon and due to confidentiality and stigma, the diagnosis might not be readily shared with the GP. One study of PLWHIV using CPRD database had a cohort of just under 3,000.(7)

Statistical analysis

For this cross sectional study, we will create a list of individuals registered with GPs on the CPRD in 2023 with a recorded diagnosis of HIV. Individuals must have diagnosis recorded on or before 2018 to give time for some types of primary health care to be delivered by 2023, e.g. cardiovascular risk assessments are normally done every five years.

We will examine specific types of care delivered: cardiovascular risk assessment for individuals over 40 years, influenza and hepatitis vaccinations, annual cervical cytology for those eligible under the screening programme. We will examine if ARTs were documented in their records for drug interactions. Each outcome will be presented as a proportion achieved out of a denominator of eligible PLWHIV.

Using regression analysis, we will measure the proportion of PLWHIV who had documented processes of care and ARTs in the records. This will be broken down by UK regions, as well as different demographic groups to examine the equity of care received. We will report descriptive statistics such as proportions for each indicator, at the UK level and at the regional level.

Workstream 3 – Co-designing a model of care (months 2 to 16, led by ANa/THI/RM/JH/SD)

Research question 3 – What does a “good” model of primary health care for PLWHIV look like?

Using a Person-Based Approach (PBA), we will co-design a model of primary health care in general practice for a pilot study. There will be two phases (see diagram) to the co-design: planning and design.(30) For the planning phase, we will identify key issues from the scoping review and CPRD analysis to discuss with the stakeholder group to co-create guiding principles.(30) These will inform the stakeholder group co-design of a model of primary care in general practice.

We will run three focus groups of up to two hours each: two face-to-face meeting for PLWHIV (one in London, one outside London, and option to join online) and online meeting for health professionals (general practice and HIV clinic staff) for their opinion on the proposed intervention. The constituents of these focus groups will be different to that of the stakeholder group to offer a range of views. We will make extra efforts to find representation from underserved PLWHIV demographics. The findings

from the focus group discussions will be fed back to the stakeholder group for further model refinement. The groups will be run by AN/THI/RM with an external facilitator. Reimbursements will be offered to all participants including expenses. Analysis will be done by RM/ANa/THI/JH.

Expected outcomes and Dissemination

Scoping review – months 2 to 9

We will identify up-to-date studies on models of care, care experiences, and map out primary health care delivered by HIV services. These would contribute to further evidence base and the co-design in workstream 3.

Analysis of primary care database - months 1 to 11

The findings will identify the engagement between PLWHIV and their GPs, and what more could be achievable and feasible for general practice to deliver.

We will identify who isn't accessing or receiving care from GPs. This might reflect the intersectionality of stigma from HIV, racism and/or sexism. Discussions at the co-design phase will include suggestions to overcome such barriers.

Non-engagement could be interpreted differently. A healthy male PLWHIV with stable disease might receive regular cardiovascular assessment in their HIV clinic, and annual influenza vaccinations from a pharmacist, so might not feel he needs GP care. Further research using qualitative studies would help to understand this better.

Codesign of a model of care - months 1-11. Focus group months 11 to 13. Refinement: month 16.

We anticipate a model of care that would be appropriate and acceptable to all stakeholders. This might include education and training for primary care providers to improve confidentiality, care navigation, improving two-way communication between GPs and HIV clinics and better access for underserved groups.

Within 18 months, we will have a model of care ready for piloting locally in Bristol, North Somerset and South Gloucester (BNSSG) area. We aim to secure a further grant (e.g. NIHR Programme Grant for Applied Research) for a feasibility study over 24 months. This model will be further developed after evaluation to be tested for effectiveness, cost-effectiveness and appropriateness in a cluster randomised trial across the UK over 36 months. We aim to use the PBA process and have the same stakeholders to give it continuity.

Our dissemination strategy will target professionals, policy makers and the public. We will publish outputs in high impact journals and present at conferences. We will work with our PPI and communications team to write news stories to share via social media and write press releases e.g. for newspapers and TV, and to engage with PLWHIV via peer networks.

We hope this project will create impact to improve the health and standardise the care of PLWHIV at scale, using a model that is acceptable to PLWHIV, HIV specialists and GPs, that would eventually be adopted in clinical guidelines.

Ethics

CPRD has broad National Research Ethics Service Committee (NRES) ethics approval for purely observational research. We will apply for University of Bristol ethics approval for the focus group studies that include healthcare professionals and PLWHIV recruited via peer networks.

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Uploads

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Year				25										26							
Month	**-3**	**-2**	**-1**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**
Month / Task/Milestone	**Dec**	**Jan**	**Feb**	**Mar**	**Apr**	**May**	**Jun**	**Jul**	**Aug**	**Sep**	**Oct**	**Nov**	**Dec**	**Jan**	**Feb**	**Mar**	**Apr**	**May**	**Jun**	**Jul**	**Aug**
Project duration																					
Research steering group																					
Recruit SRA																					
Work Programms																					
1 - scoping review																					
Manuscript write/rev/submit																					
2 - CPRD																					
Apply for approval eRAP																					
Data extraction and clean																					
Analysis																					
Manuscript write/rev/submit																					
3 - Co-Design																					
Stakeholder group																					
Focus groups																					
Manuscript write/rev/submit																					
RfPB preprations																					
RfPB next round deadlines																					
Conference (BHIVA)																					